

Production Turnover Report

Date: March 29th,2026

Business Department: GE Accounts Receivable Department

Project Name: Predictive Statistical Model for Loan Amounts

Project Description: This predictive model implements a linear regression model which was trained on a small subset of loan data from the GE Accounts Receivable Department. It takes in customer credit history and credit wellness then attempts to predict the loan amount to be made available to the customer.

Model Baselines:

Metric	Train	Test
RMSE	7519.32	7663.22
MAPE	57.40	55.73
R2	0.29	0.25

Model Performance:

The model achieves an R2 of between 0.25 and 0.29 across the training and test sets. This result indicates that the current feature set explains only a moderate part of the variance in loan amounts. The MAPE of 57% and 55% indicate a not insignificant amount of prediction error.

With this level of performance, this model should only be used as a support tool alongside human loan servicers and not as a sole decision maker.

The model's performance shouldn't change much when moving towards production. The model includes a data preprocessing pipeline which ensures that the variables used in making predictions match the ones that were used in training. The only thing that needs to be ensured is that the number of variables are not reduced in production.

Comments: The current source dataset should be examined again and potentially re-evaluated before production. This set of variables does provide insight into a customer's trustworthiness and potential to repay a loan; however, it does not examine key business-level context regarding a loan. In many lending scenarios, loans are issued for a specific need. For example, a business loan may require financing to purchase specific factory equipment needed to increase production or specific parts and materials to extend the life of a machine they already run. In these cases, the loan amount is largely determined by the cost of the business objective, rather than their credit profile alone. Contextual information is currently not a part of the data source; there is no business insight into why these loans are being serviced. The absence of this information limits the model's ability to capture a meaningful portion of the decision making in loan service.

Additionally, the dataset seems to reflect human influence in the recorded loan amounts. There are notable concentrations of values around round numbers such as 5,000, 10,000 and 25,000. This suggests that loan amounts may have been rounded or suggested by human judgement. This pattern changes the distributions of the data and limits the accuracy of a fit linear regression. It changes a continuous plane into one that is more categorical in nature. With

this information in mind, a different model like a logistic regression or decision tree might be fitted for better results in future iterations.